WORKED SOLUTION

STEP II 2005, Question 5

Pure • Incircle and Circumcircle

STEP II 2005, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q5

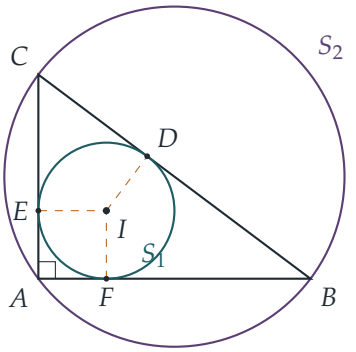
The key idea

The two tangents to a circle from an external point have equal length. Applied at each vertex, this splits every side of the triangle into two tangent lengths, and at the right angle the two tangent lengths are exactly r. That gives the formula for r almost at once. For the area ratio, the right angle tells us the hypotenuse is a diameter of S_2, and Pythagoras lets us write everything in terms of $b+c$ and a alone, which is why q appears.

Solution

The inradius

Let S_1 have centre I and touch BC, CA, AB at D, E, F respectively.



A radius meets a tangent at right angles, so $IE \perp AC$ and $IF \perp AB$. Together with the right angle at A, this makes $AEIF$ a rectangle with two adjacent sides $IE = IF = r$, so it is a square and

$$AE = AF = r.$$

Tangents from an external point are equal, so $BF = BD$ and $CE = CD$. Hence

$$c = AF + FB = r + BD, \qquad b = AE + EC = r + CD, \qquad a = BD + DC.$$

Adding the first two and subtracting the third,

$$b + c - a = 2r + BD + CD - (BD + DC) = 2r,$$

as required.

The area ratio

Since $\angle BAC = 90°$, the angle in a semicircle theorem (in reverse) says BC is a diameter of the circumcircle S_2. So S_2 has radius $\frac{1}{2}a$ and area $\frac{1}{4}\pi a^2$.

The region between S_1 and the triangle is the triangle with the disc of S_1 removed, so its area is $\frac{1}{2}bc - \pi r^2$. Therefore

$$R = \frac{\frac{1}{2}bc - \pi r^2}{\frac{1}{4}\pi a^2}, \qquad \text{so} \qquad \pi R = \frac{2bc - 4\pi r^2}{a^2} = \frac{2bc}{a^2} - \pi\,\frac{(b + c - a)^2}{a^2},$$

using $2r = b + c - a$.

Ask yourself How do we turn $2bc/a^2$ into something involving only q?

Pythagoras gives $b^2 + c^2 = a^2$, so $2bc = (b + c)^2 - (b^2 + c^2) = (b + c)^2 - a^2$. Dividing by a^2,

$$\frac{2bc}{a^2} = q^2 - 1, \qquad \frac{(b + c - a)^2}{a^2} = \left(\frac{b + c}{a} - 1\right)^2 = (q - 1)^2.$$

Hence

$$\begin{aligned} \pi R &= q^2 - 1 - \pi(q^2 - 2q + 1) \\ &= -(\pi - 1)q^2 + 2\pi q - (\pi + 1), \end{aligned}$$

which is the required result.

The bound on R

The right-hand side is a quadratic in q with negative leading coefficient $-(\pi - 1)$, so we complete the square:

$$\begin{aligned} \pi R &= -(\pi - 1)\left(q^2 - \frac{2\pi}{\pi - 1}\,q\right) - (\pi + 1) \\ &= -(\pi - 1)\left(q - \frac{\pi}{\pi - 1}\right)^2 + \frac{\pi^2}{\pi - 1} - (\pi + 1) \\ &= -(\pi - 1)\left(q - \frac{\pi}{\pi - 1}\right)^2 + \frac{\pi^2 - (\pi^2 - 1)}{\pi - 1} \\ &= \frac{1}{\pi - 1} - (\pi - 1)\left(q - \frac{\pi}{\pi - 1}\right)^2. \end{aligned}$$

Since $\pi - 1 > 0$, the squared term is subtracted, so $\pi R \le \dfrac{1}{\pi - 1}$ for every real q, and dividing by $\pi > 0$,

$$R \le \frac{1}{\pi(\pi - 1)}.$$

This is what we were asked to deduce.

It is worth noticing that the bound is not actually attained. Equality would need $q = \pi/(\pi - 1) \approx 1.467$, but for a right-angled triangle $(b + c)^2 = a^2 + 2bc \le a^2 + (b^2 + c^2) = 2a^2$ (by AM–GM, $2bc \le b^2 + c^2$), so $q \le \sqrt{2} \approx 1.414$. The largest possible R comes from the isosceles triangle, $q = \sqrt{2}$, giving $R = (2\sqrt{2}\,\pi - 3\pi + 1)/\pi \approx 0.1467$, just below the bound $1/(\pi(\pi - 1)) \approx 0.1486$.

Checks

For the 3–4–5 triangle ($b = 4$, $c = 3$, $a = 5$), the formula gives $r = 1$, and indeed the area 6 divided by the semi-perimeter 6 is 1, the standard formula $r = \text{area}/s$. Then $R = (6 - \pi)/(\frac{25}{4}\pi) \approx 0.14558$, and $q = 1.4$ in the quadratic gives the same value. A random test of 10^5 right-angled triangles confirmed the identity for πR every time and found no R above 0.14674, consistent with the bound 0.14863.

Common slips

- Forgetting that the radius of S_2 is $\frac{1}{2}a$, not a: the hypotenuse is the *diameter*.
- Trying to expand $2bc/a^2$ directly instead of using $b^2 + c^2 = a^2$ to write $2bc = (b + c)^2 - a^2$.
- Claiming the bound is attained without checking that $q = \pi/(\pi - 1)$ is achievable; it is not, though the question only asks for the inequality.

Beyond STEP

The incircle and circumcircle of a general triangle are tied together by Euler's formula $d^2 = R_2(R_2 - 2r)$ for the distance d between their centres, where R_2 is the circumradius; it implies $R_2 \ge 2r$, with equality only for the equilateral triangle. Results like this, and the idea of optimising a ratio over a family of shapes, lead into the study of geometric inequalities. Coxeter and Greitzer's *Geometry Revisited* is a beautiful next step, and Siklos's *Advanced Problems in Mathematics* (free, CC BY) has more STEP-style practice.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.